# Proving Data Structures Correct: Balanced Search Trees

Computer-Assisted Theorem Proving (7CCSACTP)

Mohammad Abdulaziz

# Search Time Depends on the Height

Week 4: `isin` follows one path from the root

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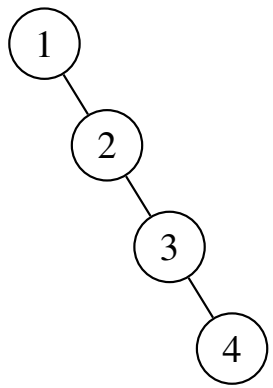
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$\Rightarrow$ search takes time $O(h(t))$, where $h(t)$ is the height of $t$

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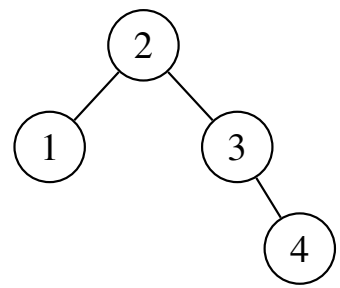
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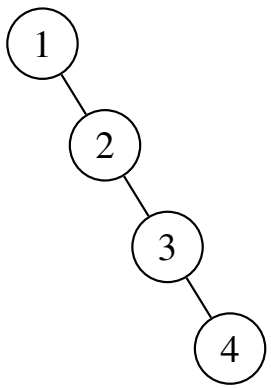

`ins` 2, 1, 3, 4: height 3

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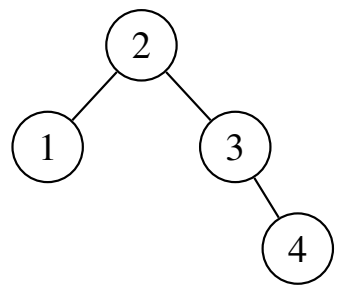
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Worst case: $h(t) = |t|$

Best case: $h(t) \approx \log_2 |t|$

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*Balanced* search trees guarantee $h(t) = O(\log |t|)$

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Examples: AVL trees, red-black trees, 2-3 trees, B-trees, ...

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